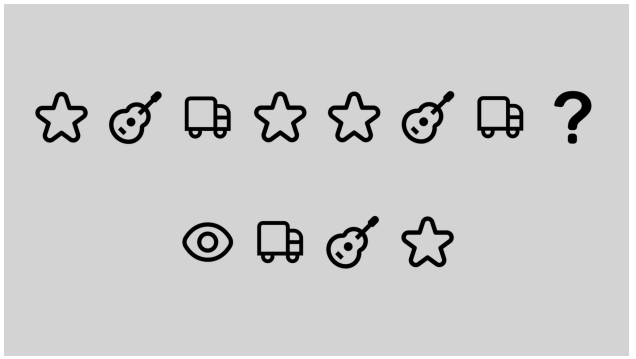
Abstract pattern reasoning in humans and language models

A common task links behavior, model representations and fixation-locked EEG

A Comparing human and model representations

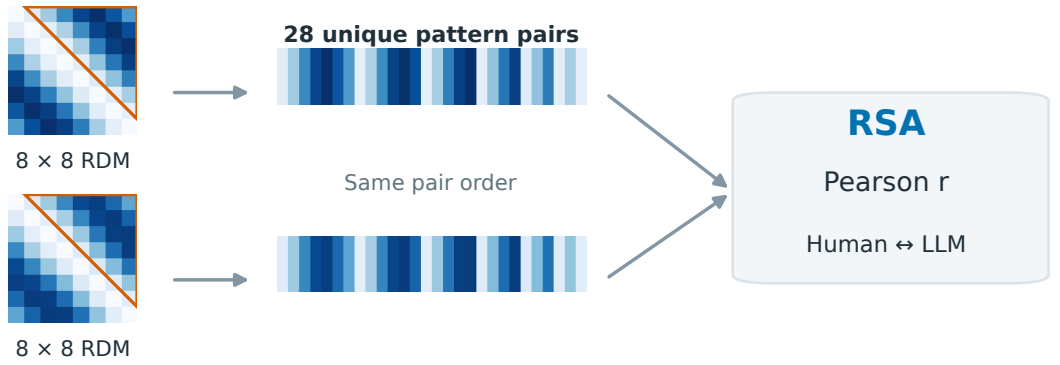


HUMAN FRP

Fixation-locked EEG → trial averages
Regional pattern means / session
RDMs: session → participant → group

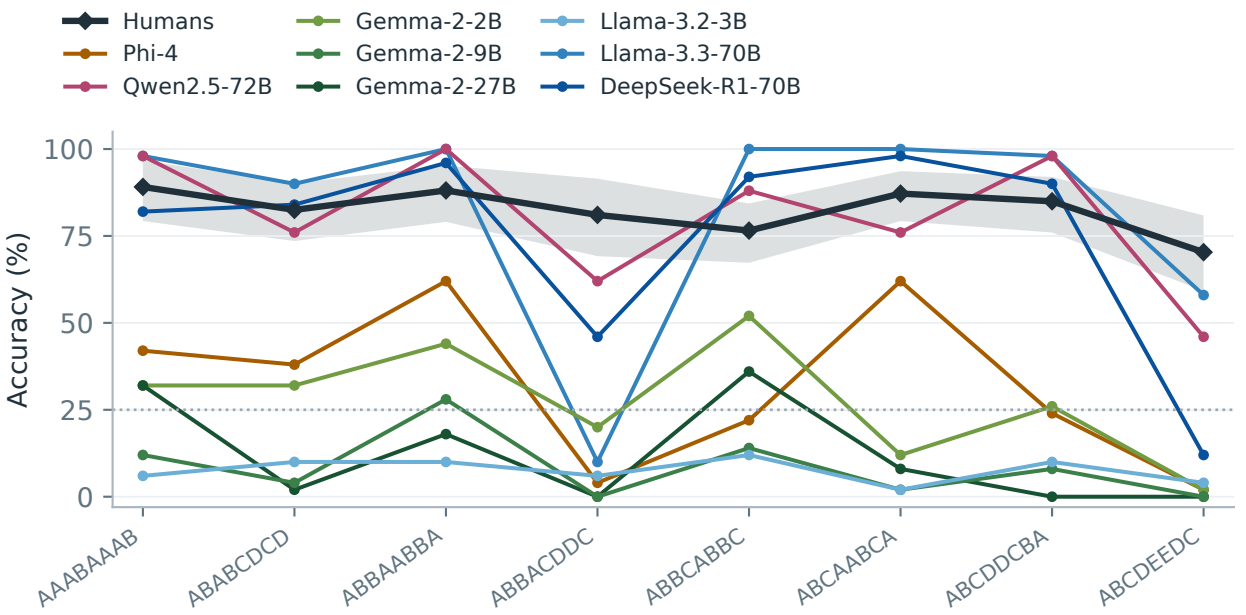
LLM REPRESENTATIONS

Sequence-token activations
Task-selected layer (no EEG used)
Average activations within pattern



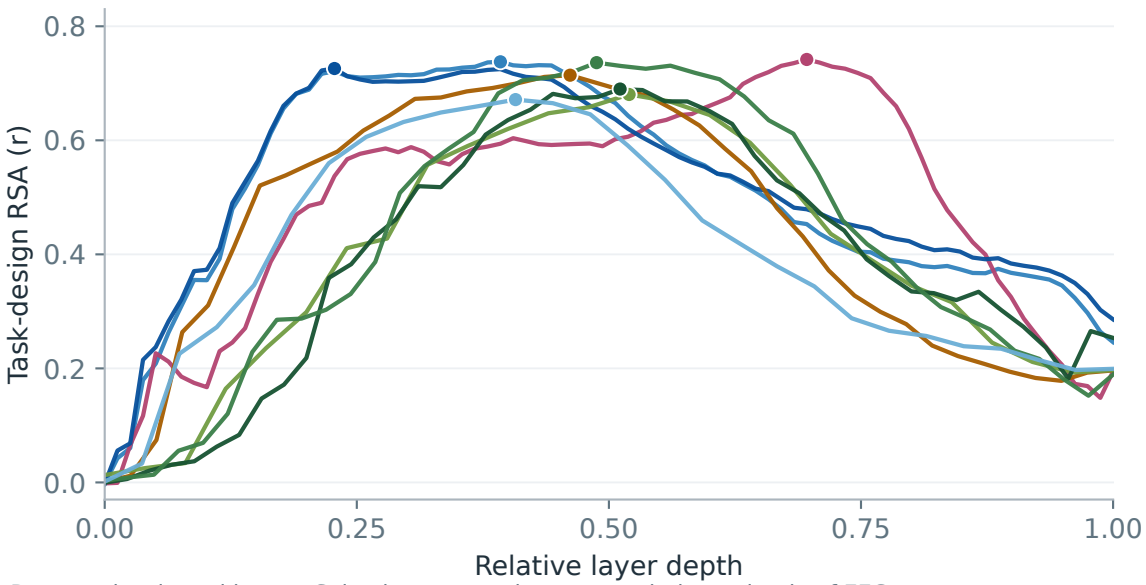
RDM distance = 1 − r. Matrices are schematic; RSA compares corresponding upper-triangle entries.

B Accuracy across abstract patterns



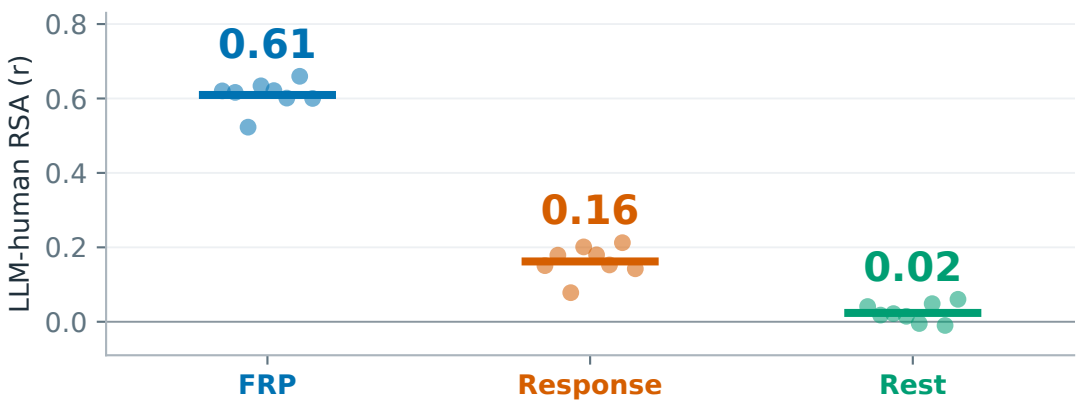
Human band: 95% bootstrap CI (N = 25). Dotted line: 25% chance.

C Task structure emerges inside the models



Dots mark selected layers. Selection uses task structure, independently of EEG.

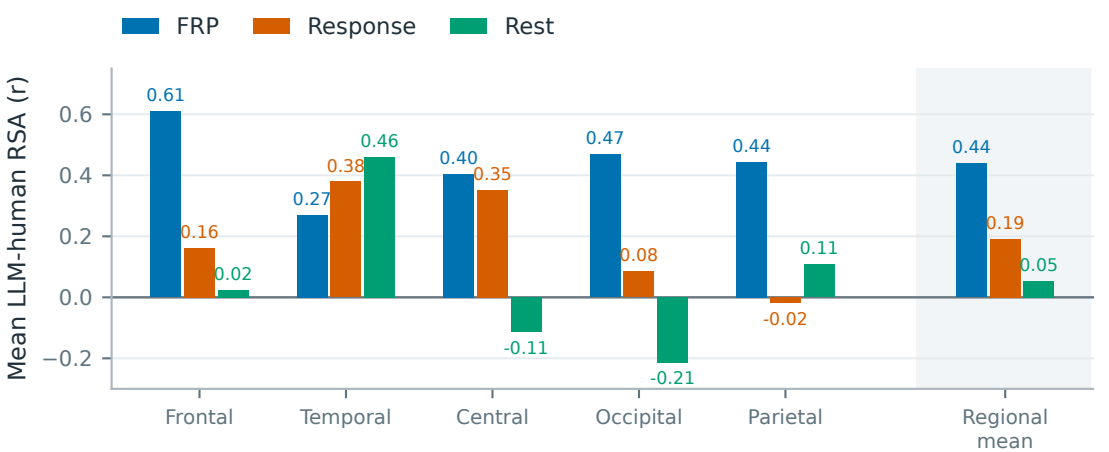
D Frontal correspondence is strongest for FRP



Dots: eight models. Horizontal marks: means. Matched N = 24.

Frontal FRP pattern-label test: p = .010; Holm-adjusted p = .030.

E Regional context matters



Bars average eight models; shaded group: mean across the five regions.